Physical chemistry

25 questions: 8 multiple choice, 17 structured.

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1 · Part 1 Paper 1 & 2 Energetics [1]

Which reaction is endothermic?

A the combustion of methane

B the thermal decomposition of calcium carbonate

C the neutralisation of hydrochloric acid by sodium hydroxide solution

D the reaction of sodium with water

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2 · Part 1 Paper 1 & 2 Energetics [1]

In a calorimetry experiment, 50 g of water is heated from 18.0 °C to 34.0 °C.

The specific heat capacity of water is 4.2 J/g/°C.

What is the heat energy taken in by the water?

A 7.14 kJ

B 3360 kJ

C 0.80 kJ

D 3.36 kJ

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3 · Part 1 Paper 2 only Energetics [1]

Which change is exothermic?

A $H(g) + H(g) \rightarrow H_2(g)$

B $Cl_2(g) \rightarrow Cl(g) + Cl(g)$

C $H_2O(l) \rightarrow H_2O(g)$

D $O_2(g) \rightarrow O(g) + O(g)$

4 · Part 1

Paper 1 & 2 Energetics [7]

Zinc displaces copper from copper(II) sulfate solution. The reaction gives out heat energy.

$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$

Sam wants to measure the temperature rise for this reaction. Sam has zinc powder, copper(II) sulfate solution, a polystyrene cup with a lid, a thermometer, a measuring cylinder and a balance.

(a) Describe how Sam could carry out the experiment to find the temperature rise. [4]

(b) Give a reason why Sam uses a polystyrene cup with a lid. [1]

(c) Suggest why using zinc powder, rather than one large piece of zinc, gives a more accurate value for the temperature rise. [2]

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5 · Part 1 Paper 1 & 2 Energetics [7]

Leila mixes some dilute hydrochloric acid with some sodium hydroxide solution in a polystyrene cup.

$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$

The hydrochloric acid contains 0.0500 mol of HCl and the sodium hydroxide solution contains 0.0500 mol of NaOH. Both solutions start at 19.0 °C. The highest temperature reached by the mixture is 32.5 °C.

The mixture has a mass of 50.0 g and a specific heat capacity of 4.2 J/g/°C.

(a) Calculate the heat energy change, Q, in J. [2]

(b) Calculate the molar enthalpy change, ΔH, for the reaction in kJ per mole of HCl. Include a sign in your answer. [3]

(c) Leila repeats the experiment with the same amounts of HCl and NaOH, but each is dissolved in more water, so the mixture has a mass of 100 g. Both solutions again start at 19.0 °C. Predict the temperature rise in this experiment. Explain your answer. [2]

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6 · Part 1

Paper 1 & 2 Energetics [9]

Omar uses a spirit burner containing hexane, C_6H_{14}, to heat 200 g of water in a copper can. He weighs the burner before and after heating and measures the temperature of the water.

Measurement	Value
mass of burner at start	152.37 g
mass of burner at end	151.51 g
temperature of water at start	18.5 °C
temperature of water at end	43.5 °C

The specific heat capacity of water is 4.2 J/g/°C. Use the Periodic Table.

(a) Calculate the amount, in moles, of hexane burned. [3]

(b) Calculate the heat energy change, Q, in kJ. [2]

(c) Calculate the molar enthalpy change, ΔH, for the combustion of hexane in kJ/mol. Include a sign in your answer. [2]

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(d) The data book value for the molar enthalpy change of combustion of hexane is −4160 kJ/mol. Explain why Omar's value is much less negative than the data book value. [2]

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7 · Part 1

Paper 1 & 2 | Energetics | [8]

Chen investigates the temperature change when four salts, W, X, Y and Z, dissolve in water. For each salt, he adds 5.0 g of the salt to 50 cm^3 of water in a polystyrene cup, stirs, and records the starting temperature and the final temperature.

(a) The table shows Chen's results. Complete the table by calculating the temperature change for each salt. Use + for a rise and – for a fall.

Salt	Starting temperature in °C	Final temperature in °C	Temperature change in °C
W	20.5	14.0	
X	20.0	27.5	
Y	21.0	19.0	
Z	20.5	33.0	

[2]

(b) Identify the salts that dissolve endothermically. Give a reason for your answer. [2]

(c) Self-heating food cans contain a substance that gives out heat when it dissolves in water. Identify which salt would be most suitable for a self-heating can. Give a reason for your answer. [2]

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(d) State two variables that Chen kept the same so that his comparison is fair. [2]

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8 · Part 1

Paper 2 only

Energetics

[6]

Hydrazine, N_2H_4, is used as a rocket fuel. It reacts with oxygen.

$N_2H_4(g) + O_2(g) \rightarrow N_2(g) + 2H_2O(g)$

Each hydrazine molecule, $H_2N–NH_2$, contains four N–H bonds and one N–N bond. Each oxygen molecule contains one O=O bond, each nitrogen molecule contains one N≡N bond and each water molecule contains two O–H bonds.

The table gives some bond energies.

Bond	Bond energy in kJ/mol
N–H	391
N–N	158
O=O	498
N≡N	945
O–H	464

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(a) Calculate the enthalpy change, ΔH, for the reaction. Include a sign in your answer. [4]

(b) Explain, in terms of bond breaking and bond making, why this reaction is exothermic. [2]

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9 · Part 1

Paper 2 only

Energetics

[6]

Hydrogen reacts with bromine to form hydrogen bromide.

$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$ $\Delta H = -103$ kJ/mol

The table gives some bond energies.

Bond	Bond energy in kJ/mol
H–H	436
Br–Br	193
F–F	158
H–F	568

(a) Calculate the bond energy of the H–Br bond. [3]

(b) Hydrogen also reacts with fluorine: $H_2(g) + F_2(g) \rightarrow 2HF(g)$. Calculate the enthalpy change, ΔH, for this reaction and deduce whether the reaction is exothermic or endothermic. [3]

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10 · Part 1 Paper 1 & 2 Rates of reaction [1]

Zinc powder reacts faster with dilute sulfuric acid than the same mass of zinc granules.

Which statement explains why?

A The zinc powder lowers the activation energy of the reaction.

B The particles in zinc powder have more energy than the particles in zinc granules.

C The zinc powder has a larger surface area, so collisions with acid particles are more frequent.

D The zinc powder increases the concentration of the sulfuric acid.

11 · Part 1 Paper 1 & 2 Rates of reaction [1]

Which statement describes a catalyst?

A It increases the rate of a reaction and is chemically unchanged at the end of the reaction.

B It increases the rate of a reaction and is used up during the reaction.

C It increases the rate of a reaction and increases the mass of product made.

D It increases the rate of a reaction by raising the temperature of the reaction mixture.

12 · Part 1 Paper 1 & 2 Rates of reaction [1]

Kofi reacts marble chips with dilute hydrochloric acid and measures the total volume of carbon dioxide produced. The hydrochloric acid is in excess in every experiment.

Which change increases the rate of reaction but does not change the total volume of carbon dioxide produced?

A using the same mass of larger marble chips

B using the same mass of powdered marble

C using twice the mass of marble chips of the same size

D using hydrochloric acid of a lower concentration

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13 · Part 1

Paper 1 & 2 | Rates of reaction | [8]

Aisha investigates the reaction between marble chips and dilute hydrochloric acid.

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

She puts a conical flask containing the acid on a balance, adds the marble chips and places a loose plug of cotton wool in the neck of the flask. She records the total loss in mass every 30 s. The marble chips are in excess.

Time in s	**0**	**30**	**60**	**90**	**120**	**150**	**180**	**210**
Total loss in mass in g	0.00	0.42	0.74	0.96	1.10	1.18	1.20	1.20

(a) Give a reason why the mass of the flask and its contents decreases. [1]

(b) Calculate the mean rate of reaction in the first 60 s, in g/s. [2]

(c) Explain, in terms of particles, why the rate of reaction decreases as the reaction continues. [2]

(d) Give a reason why the total loss in mass does not change after 180 s. [1]

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(e) Describe how Aisha could change her method to investigate the effect of the concentration of the hydrochloric acid on the rate of this reaction. [2]

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14 · Part 1

Paper 1 & 2 | Rates of reaction | [8]

Hydrogen peroxide solution decomposes slowly to form water and oxygen gas.

Tom adds 0.5 g of a solid to 25 cm^3 of hydrogen peroxide solution and collects the oxygen in a gas syringe. He measures the volume of oxygen collected in 60 s. He repeats the experiment with different solids, using fresh hydrogen peroxide solution each time.

Solid added	Volume of oxygen collected in 60 s in cm^3
no solid	1
manganese(IV) oxide	52
copper(II) oxide	14
zinc oxide	9
sand	1

(a) Complete the balanced equation for the decomposition of hydrogen peroxide. Include state symbols.

....H_2O_2(aq) → ....H_2O(....) + O_2(....) [2]

(b) Identify the most effective catalyst for this reaction. Give a reason for your answer. [2]

(c) Identify one solid in the table that does not act as a catalyst for this reaction. Give a reason for your answer. [1]

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(d) Describe how Tom could show that the mass of manganese(IV) oxide is unchanged at the end of the reaction. [3]

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15 · Part 1

Paper 1 & 2 Rates of reaction [6]

Carbon monoxide gas reacts with nitrogen dioxide gas in a closed container.

$CO(g) + NO_2(g) \rightarrow CO_2(g) + NO(g)$

(a) Explain, in terms of particle collisions, why increasing the pressure of the gas mixture increases the rate of this reaction. [2]

(b) Explain, in terms of particle collisions, why increasing the temperature increases the rate of this reaction. [3]

(c) The same amounts of the two gases are put into a larger container at the same temperature. Predict the effect on the rate of reaction. [1]

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16 · Part 1

Paper 1 & 2 Rates of reaction [7]

Sodium thiosulfate solution reacts with dilute hydrochloric acid. One product is sulfur, which forms as a fine yellow precipitate, so the mixture slowly turns cloudy.

Grace places a conical flask on a piece of paper marked with a cross. She times how long it takes for the cross to disappear from view when the solutions are mixed in the flask. She does this at five different temperatures.

Temperature in °C	**20**	**30**	**40**	**50**	**60**
Time for cross to disappear in s	124	64	33	17	9

(a) Describe how Grace could carry out the experiment to obtain these results. [4]

(b) Describe the effect of temperature on the rate of this reaction. Use data from the table in your answer. [2]

(c) Suggest why the result at 60 °C is likely to be the least accurate. [1]

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17 · Part 1

Paper 1 & 2 Rates of reaction [6]

Ravi adds a piece of magnesium ribbon to dilute hydrochloric acid and collects the hydrogen in a gas syringe. The acid is in excess.

$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

Time in s	**0**	**10**	**20**	**30**	**40**	**50**	**60**	**80**	**100**	**120**
Volume of hydrogen in cm^3	0	22	38	50	58	64	68	71	72	72

(a) Calculate the mean rate of reaction in the first 20 s, in cm^3/s. [2]

(b) Ravi plots a graph of these results and draws a tangent to the curve at 40 s. The tangent passes through the points (10 s, 37 cm^3) and (70 s, 79 cm^3). Calculate the rate of reaction at 40 s. Give the unit. [2]

(c) Ravi repeats the experiment with an identical piece of magnesium ribbon and the same volume of a more concentrated hydrochloric acid. The acid is still in excess. Predict the final volume of hydrogen collected. Give a reason for your answer. [2]

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18 · Part 1 Paper 1 & 2 Rates of reaction [5]

Dinitrogen oxide gas decomposes when it is heated.

$2N_2O(g) \rightarrow 2N_2(g) + O_2(g)$

The table gives the activation energy of the reaction with and without a platinum catalyst.

Condition	Activation energy in kJ/mol
without a catalyst	250
with a platinum catalyst	135

(a) State what is meant by the term activation energy. [1]

(b) Explain how the platinum catalyst increases the rate of this reaction. [3]

(c) Suggest one reason why using a catalyst can reduce the cost of running an industrial process. [1]

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19 · Part 1

Paper 1 & 2 | Reversible reactions and equilibria | [1]

Hydrated copper(II) sulfate is heated in a test tube. The solid left in the tube is cooled and then a few drops of water are added to it.

Which row shows the colour changes?

	Colour change on heating	Colour change when water is added
A	white to blue	blue to white
B	blue to white	blue to white
C	white to blue	white to blue
D	blue to white	white to blue

20 · Part 1

Paper 2 only | Reversible reactions and equilibria | [1]

Carbon monoxide and hydrogen react to form methanol in a reversible reaction.

$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$ $\Delta H = -91$ kJ/mol

Which row shows two changes that would both shift the position of equilibrium to the right?

	Temperature	Pressure
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

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21 · Part 1

Paper 1 & 2 Reversible reactions and equilibria [7]

When solid ammonium chloride is heated, it breaks down into ammonia gas and hydrogen chloride gas. The reaction is reversible.

(a) State what the symbol ⇌ shows in a chemical equation. [1]

(b) Complete the equation for this reaction by giving the formulae and state symbols of the two products.

$NH_4Cl(s) \rightleftharpoons$ ..... + ... [2]

(c) A small amount of ammonium chloride is heated at the bottom of a test tube. The top of the test tube stays cool. Describe what is seen at the bottom of the tube and near the top of the tube. [2]

(d) Explain why a solid forms near the top of the tube. [2]

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22 · Part 1

Paper 2 only

Reversible reactions and equilibria

[8]

Dinitrogen tetroxide, N_2O_4, is a colourless gas. Nitrogen dioxide, NO_2, is a brown gas. A mixture of the two gases is sealed in a gas syringe at room temperature and reaches dynamic equilibrium.

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

(a) Give a reason why the gases must be in a sealed container for dynamic equilibrium to be reached. [1]

(b) After a few minutes the colour of the mixture stops changing. Explain why the colour stays the same even though both the forward and the reverse reactions are still taking place. [2]

(c) The syringe is clamped so that its volume cannot change and is then placed in hot water. The mixture becomes a darker brown. Deduce whether the forward reaction is exothermic or endothermic. Explain your answer. [3]

(d) The syringe is unclamped and returned to room temperature. The plunger is then pushed in, which increases the pressure. State and explain the direction in which the position of equilibrium moves. [2]

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23 · Part 1

Paper 2 only

Reversible reactions and equilibria

[7]

Sulfur trioxide is made by reacting sulfur dioxide with oxygen.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -196$ kJ/mol

The industrial process uses a temperature of about 450 °C, a pressure of about 2 atm and a vanadium(V) oxide catalyst.

(a) Explain the effect of increasing the pressure on the equilibrium yield of sulfur trioxide. [2]

(b) Explain why the vanadium(V) oxide catalyst does not change the equilibrium yield of sulfur trioxide. [2]

(c) A lower temperature would give a higher equilibrium yield of sulfur trioxide. Explain why a lower temperature gives a higher yield, and why a temperature of about 450 °C is used instead. [3]

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24 · Part 1

Paper 2 only

Reversible reactions and equilibria

[5]

Hydrogen can be made by reacting methane with steam in a reversible reaction.

$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$

The tables show the percentage of methane converted at equilibrium under different conditions.

Results at a pressure of 10 atm:

Temperature in °C	Percentage of methane converted
600	30
700	55
800	80
900	93

Results at a temperature of 800 °C:

Pressure in atm	Percentage of methane converted
1	97
10	80
30	60

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(a) Deduce whether the forward reaction is exothermic or endothermic. Explain your answer using the data. [2]

(b) Explain why the percentage of methane converted decreases as the pressure increases. [2]

(c) In industry, this reaction is often carried out at about 30 atm, even though less methane is converted at this pressure. Suggest one reason for using this pressure. [1]

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25 · Part 1

Paper 2 only

Reversible reactions and equilibria

[6]

Hydrogen and iodine gases are mixed in a sealed container at a constant high temperature. They react to form hydrogen iodide in a reversible reaction.

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

The table shows the amounts of each gas in the container over time.

Time in min	Amount of H_2 in mol	Amount of I_2 in mol	Amount of HI in mol
0	1.00	1.00	0.00
1	0.60	0.60	0.80
2	0.40	0.40	1.20
3	0.30	0.30	1.40
4	0.22	0.22	1.56
5	0.22	0.22	1.56
6	0.22	0.22	1.56

(a) Identify the earliest time in the table at which the mixture is at dynamic equilibrium. Give a reason for your answer. [2]

(b) Describe and explain how the rates of the forward and reverse reactions change between 0 min and the time when equilibrium is reached. [4]

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